

Instructions:

- You may use any calculator except a TI-nspire.
- Put away all phones, laptops, and course materials.
- Show all of your work, and use correct notation. Failure to do these things will result in a deduction of points.
- You have **1 hour** to complete this exam.
- You've got this—I believe in you!

1. (22 pts.) Suppose $Y_1, Y_2, \dots, Y_6 \stackrel{i.i.d.}{\sim} N(0, 1)$.

(a.) (4 pts.) What is the distribution of $\bar{Y} = \frac{1}{6} \sum_{i=1}^6 Y_i$?

(b.) (4 pts.) What is the distribution of $\sum_{i=1}^6 Y_i^2$?

(c.) (10 pts.) Find an interval that will contain $S^2 = \frac{1}{5} \sum_{i=1}^6 (Y_i - \bar{Y})^2$ with probability .99.

(d.) (4 pts.) What is the distribution of $\bar{Y}/(S/\sqrt{6})$?

2. (26 pts.) Consider $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} \text{Uniform}(\theta, \theta + 6)$. Suppose we wish to estimate θ . Consider the estimator $\hat{\theta} = \bar{Y} + 3$.

(a.) (6 pts.) Calculate $E(\hat{\theta})$.

(b.) (6 pts.) Calculate $B(\hat{\theta})$.

(c.) (8 pts.) Calculate $MSE(\hat{\theta})$.

(d.) (6 pts.) Modify $\hat{\theta}$ to get an unbiased estimator of θ . Call this new estimator $\hat{\theta}^*$.

3. (22 pts.) Suppose that Y is a single observation from an exponential distribution with mean θ .

(a.) (10 pts.) Using the method of moment-generating functions, find the distribution of $U = \frac{Y}{2\theta}$.

(b.) (12 pts.) Using U as your pivotal quantity, find a 95% confidence interval for θ .

4. (18 pts.) A wildlife researcher is studying the population of a certain species of frogs in a large nature reserve. She randomly selects a sample of 200 frogs and finds that 34 of them have a specific skin disease.

(a.) (10 pts.) Calculate a 90% confidence interval for the true proportion of frogs in this reserve that have this skin disease. You may assume the conditions needed to construct a large-sample confidence interval are satisfied (they are ☺).

(b.) (8 pts.) Interpret your interval in context of the problem.

Choose **one** of problems 5-7 to complete. If you work on more than one problem, indicate which one you want graded; otherwise, your instructor may randomly select which one to grade.

5. (12 pts.) Prove that $MSE(\hat{\theta})$, which is equal to $E[(\hat{\theta} - \theta)^2]$, can be expressed as

$$MSE(\hat{\theta}) = V(\hat{\theta}) + [B(\hat{\theta})]^2.$$

6. (12 pts.) Suppose that $E(\hat{\theta}_1) = E(\hat{\theta}_2) = \theta$, $V(\hat{\theta}_1) = \sigma_1^2 > 0$, and $V(\hat{\theta}_2) = \sigma_2^2 > 0$. Consider the unbiased estimator $\hat{\theta}_3 = a\hat{\theta}_1 + (1 - a)\hat{\theta}_2$, where a is a constant. If $\hat{\theta}_1$ and $\hat{\theta}_2$ are independent, what value of a should be chosen to minimize the variance of $\hat{\theta}_3$? Verify that this value is, indeed, a minimizer.

7. (12 pts.) Let $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$ and $T = \frac{\bar{Y} - \mu}{S/\sqrt{n}} = \sqrt{n} \left(\frac{\bar{Y} - \mu}{S} \right)$. Prove that $T \sim t_{n-1}$.
You may reference the following fact in your proof:

Fact: Suppose $Z \sim N(0, 1)$ and $W \sim \chi_\nu^2$. Then, if Z and W are independent,

$$T = \frac{Z}{\sqrt{W/\nu}} \sim t_\nu.$$